

Example of isotope

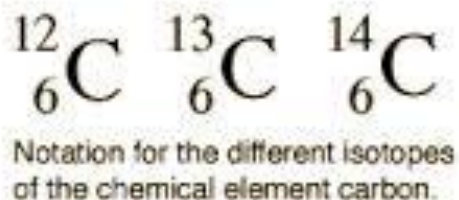
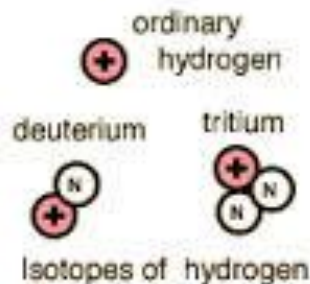
ISOTOPES OF CARBON

	Carbon-12	Carbon-13	Carbon-14
Protons	6	6	6
Neutrons	6	7	8
Electrons	6	6	6

^{16}O 15.9949 99.76% Stable	^{17}O 16.9991 0.04% Stable	^{18}O 17.9991 0.20% Stable
--	---	---

^{12}C 12.00000 98.89% Stable	^{13}C 13.00335 1.11% Stable	^{14}C 14.0 $t_{1/2} = 5715\text{yrs}$ Radioactive Cosmogenic/ anthropogenic
---	--	--

^{14}N 14.00307 99.63% Stable	^{15}N 15.0001 0.37% Stable
---	---

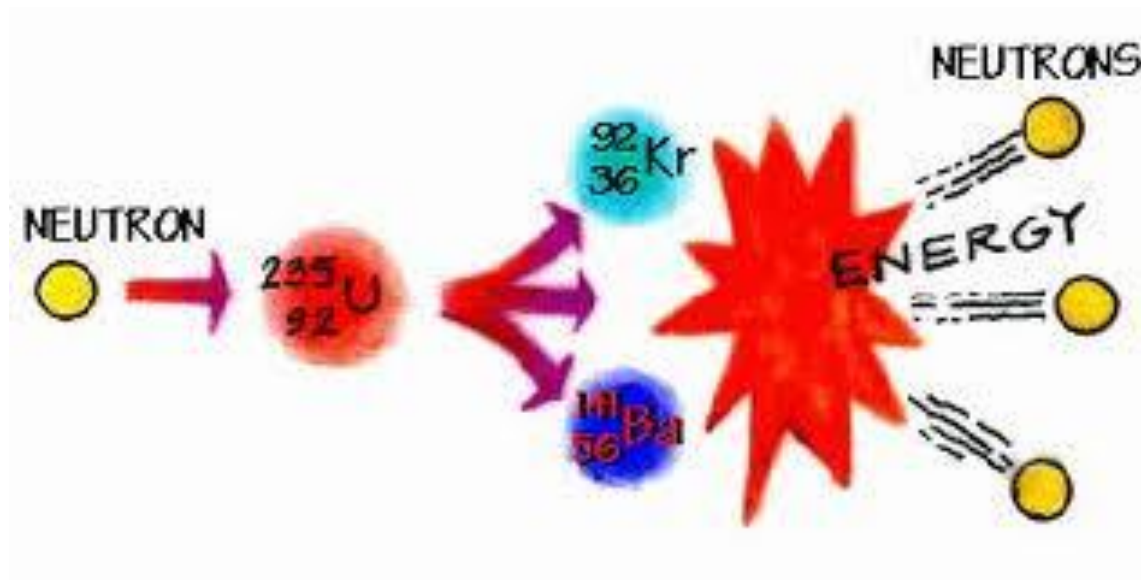


There are about 400 stable isotopes.

Problem

The isotope of uranium used to generate nuclear power is $^{235}_{92}\text{U}$.

How many protons, neutrons, and electrons does an atom of $^{235}_{92}\text{U}$ have?



Problem

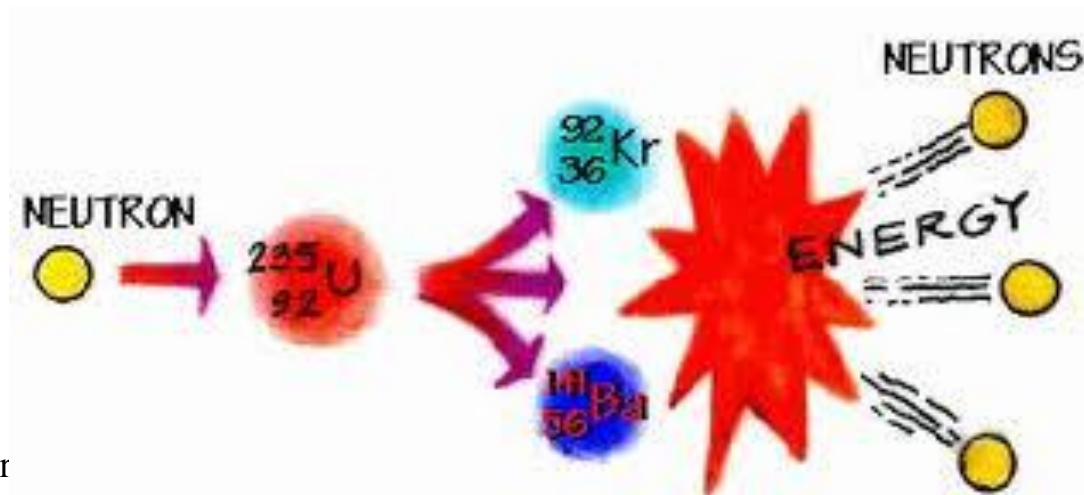
The isotope of uranium used to generate nuclear power is ${}^{235}_{92}\text{U}$.

How many protons, neutrons, and electrons does an atom of ${}^{235}_{92}\text{U}$ have?

Solve:

An atom of ${}^{235}_{92}\text{U}$ has 92 protons, 92 electrons, and

$235 - 92 = 143$ neutrons.



Molecule

A molecule is an aggregate of at least two atoms in a definite arrangement held together by chemical forces (also called chemical bonds)

A molecule may contain **atoms of the same element** or **atoms of two or more elements** joined in a fixed ratio.

Diatomic molecule: Diatomic molecule contains only two atoms.

Example: H_2 , N_2 , O_2 , Br_2 , HCl , CO

Polyatomic molecule: Molecules containing more than two atoms are called polyatomic molecules.

Example: water (H_2O) and ammonia (NH_3) are polyatomic molecules.

Ion

An ion is **an atom or a group of atoms** that has a net positive or negative charge.

1 The loss of one or more electrons from a neutral atom results in a cation, an ion with a net positive charge.

For example: sodium atom (Na) can readily lose an electron to become a sodium cation, which is represented by Na^+ .

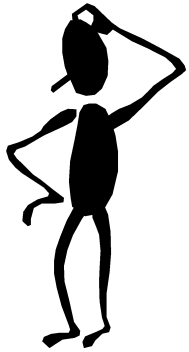
2 On the other hand, an anion is an ion whose net charge is negative due to an increase in the number of electrons. A chlorine atom (Cl), can gain an electron to become the chloride ion Cl^- .

Ions

Na Atom	Na ⁺ Ion
11 protons	11 protons
11 electrons	10 electrons

Cl Atom	Cl ⁻ Ion
17 protons	17 protons
17 electrons	18 electrons

Sodium chloride (NaCl), **ordinary table salt**, is called an **ionic compound** because it is formed from cations and anions.



Do You Understand Ions?

How many protons and electrons are in



13 protons, 10 ($13 - 3$) electrons

How many protons and electrons are in



34 protons, 36 ($34 + 2$) electrons

Ions

1 1A	2 2A											13 3A	14 4A	15 5A	16 6A	17 7A	18 8A
Li ⁺													C ⁴⁻	N ³⁻	O ²⁻	F ⁻	
Na ⁺	Mg ²⁺	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	Al ³⁺		P ³⁻	S ²⁻	Cl ⁻	
K ⁺	Ca ²⁺				Cr ²⁺ Cr ³⁺	Mn ²⁺ Mn ³⁺	Fe ²⁺ Fe ³⁺	Co ²⁺ Co ³⁺	Ni ²⁺ Ni ³⁺	Cu ⁺ Cu ²⁺	Zn ²⁺				Se ²⁻	Br ⁻	
Rb ⁺	Sr ²⁺									Ag ⁺	Cd ²⁺		Sn ²⁺ Sn ⁴⁺		Te ²⁻	I ⁻	
Cs ⁺	Ba ²⁺									Au ⁺ Au ³⁺	Hg ₂ ²⁺ Hg ²⁺		Pb ²⁺ Pb ⁴⁺				

Figure 2.10

Common monatomic ions arranged according to their positions in the periodic table. Note that the Hg_2^{2+} ion contains two atoms.

Problem

Give the number of protons, neutrons, and electrons in each of the following species:

(a) $^{195}_{79}\text{Au}$, (b) $^{197}_{79}\text{Au}$, (c) ^{18}F , (d) carbon-13.

Solution

- (a) The atomic number of Au (gold) is 79, so there are 79 protons. The mass number is 195, so the number of neutrons is $195 - 79 = 116$. The number of electrons is the same as the number of protons; that is, 79.
- (b) Here the number of protons is the same as in (a), or 79. The mass number is 197, so the number of neutrons is $197 - 79 = 118$. The number of electrons is also the same as in (a), 79. The species in (a) and (b) are chemically similar isotopes of gold.
- (c) The atomic number of F (fluorine) is 9, so there are 9 protons. The mass number is 18, so the number of neutrons is $18 - 9 = 9$. The number of electrons is the same as the number of protons; that is, 9.
- (d) Carbon-13 can also be represented as ^{13}C . The atomic number of carbon is 6, so there are $13 - 6 = 7$ neutrons. The number of electrons is 6.

The Periodic table

Alkali Metals

Noble Gases

Alkaline Earths

Halogens

Main Group

Transition Metals

Alkaline Earths						Halogens						Main Group					18	
1 1A																		8A
1 H 1.00794	2 2A	Transition Metals										13 3A	14 4A	15 5A	16 6A	17 7A	2 He 4.00260	
3 Li 6.941	4 Be 9.01218	3 3B	4 4B	5 5B	6 6B	7 7B	8 8B	9 8B	10 8B	11 1B	12 2B	5 B 10.811	6 C 12.011	7 N 14.0067	8 O 15.9994	9 F 18.9984	10 Ne 20.1797	
11 Na 22.9898	12 Mg 24.3050	21 Sc 44.9559	22 Ti 47.88	23 V 50.9415	24 Cr 51.9961	25 Mn 54.9381	26 Fe 55.847	27 Co 58.9332	28 Ni 58.693	29 Cu 63.546	30 Zn 65.39	13 Al 26.9815	14 Si 28.0855	15 P 30.9738	16 S 32.06	17 Cl 35.4527	18 Ar 39.948	
19 K 39.0983	20 Ca 40.078	39 Y 88.9059	40 Zr 91.224	41 Nb 92.9064	42 Mo 95.94	43 Tc (98)	44 Ru 101.07	45 Rh 102.906	46 Pd 106.42	47 Ag 107.868	48 Cd 112.411	31 Ga 69.723	32 Ge 72.61	33 As 74.9216	34 Se 78.96	35 Br 79.904	36 Kr 83.80	
37 Rb 85.4678	38 Sr 87.62	57 *La 138.906	72 Hf 178.49	73 Ta 180.948	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.967	80 Hg 200.59	49 In 114.818	50 Sn 118.710	51 Sb 121.757	52 Te 127.60	53 I 126.904	54 Xe 131.29	
55 Cs 132.905	56 Ba 137.327	72 Hf 178.49	73 Ta 180.948	74 W 183.84	75 Re 186.207	76 Os 190.23	77 Ir 192.22	78 Pt 195.08	79 Au 196.967	80 Hg 200.59		81 Tl 204.383	82 Pb 207.2	83 Bi 208.980	84 Po (209)	85 At (210)	86 Rn (222)	
87 Fr (223)	88 Ra 226.025	89 †Ac 227.028	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 (269)	111 (272)	112 (272)		114 (287)		116 (289)		118 (293)	
*Lanthanide series		58 Ce 140.115	59 Pr 140.908	60 Nd 144.24	61 Pm (145)	62 Sm 150.36	63 Eu 151.965	64 Gd 157.25	65 Tb 158.925	66 Dy 162.50	67 Ho 164.930	68 Er 167.26	69 Tm 168.934	70 Yb 173.04	71 Lu 174.967			
†Actinide series		90 Th 232.038	91 Pa 231.036	92 U 238.029	93 Np 237.048	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)			

Main Group

Lanthanides and Actinides

Periodic Table

- **Note:** Under normal conditions, elements in **Black** are solids (K, S); elements in **Blue** are liquids (Hg & Br₂); elements in **Red** are gases (H₂, F₂, Ar).
- *IMPORTANT Memorize:* Hydrogen, Nitrogen, Oxygen, Fluorine, Chlorine, Bromine & Iodine normally exist as diatomic molecules, & these molecules have the same names as the atoms:



hydrogen, nitrogen, oxygen, fluorine, chlorine, bromine, iodine

Mole (abbreviated, *mol*)

One mole of any substance is the amount whose mass in grams – *its molar mass*- is numerically equal to its molecular or formula weight.

Example: One mole of ethylene has a mass of 28.0 g
 One mole of HCl has a mass of 36.5 g

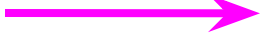
Mole: The amount of a substance whose mass in grams is numerically equal to its molecular or formula weight.

Mole

Moles tell us *how many molecules* of each reactant are needed

Grams tell us *how much mass* of each reactant is needed

Moles  Numbers of molecules

Grams  Mass

Problem

How many moles of sucrose are in a tablespoon of sugar that contains 2.85 g? (Molar mass of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, is 342)

Problem

How many moles of sucrose are in a tablespoon of sugar that contains 2.85 g? (Molar mass of sucrose, $\text{C}_{12}\text{H}_{22}\text{O}_{11}$, is 342)

Solve:

$$2.85 \text{ g sucrose} \times \frac{1 \text{ mol sucrose}}{342 \text{ g sucrose}} = 0.00833 \text{ mol sucrose}$$
$$= 8.33 \times 10^{-3} \text{ mol sucrose}$$

Note: As molecular mass of sucrose is 342 amu, 1 mol of sucrose has a mass of 342 g.

Problem

Solve:

Note: As molecular mass of sucrose is 342 amu, 1 mol of sucrose has a mass of 342 g.

$$\begin{aligned} 2.85 \text{ g sucrose} \times \frac{1 \text{ mol sucrose}}{342 \text{ g sucrose}} &= 0.00833 \text{ mol sucrose} \\ &= 8.33 \times 10^{-3} \text{ mol sucrose} \end{aligned}$$

Problem

How many grams are in 0.0626 mol of NaHCO_3 , the main ingredient
In Alka-Seltzer (antacid) tablets?

Problem

How many grams are in 0.0626 mol of NaHCO_3 , the main ingredient in Alka-Seltzer (antacid) tablets?

Solve:

$$\text{Molar mass of NaHCO}_3 = 84.0 \text{ g/mol}$$

We know,

$$\text{Mole} = \frac{\text{Mass}}{\text{Molar mass}}$$

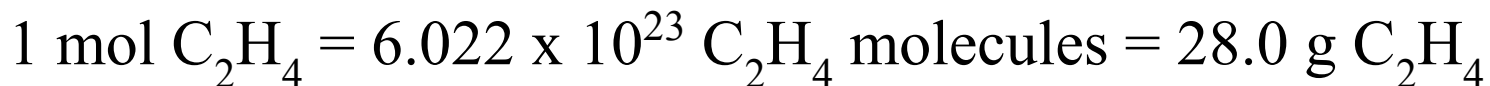
$$0.0626 \text{ mol NaHCO}_3 \times \frac{84.0 \text{ g NaHCO}_3}{1 \text{ mol NaHCO}_3} = 5.26 \text{ g NaHCO}_3$$

Avogadro's number (N_A)

How many molecules are there in a mole?

Experiments show that one mole of any substance contains 6.022×10^{23} formula units, a value called **Avogadro's number (N_A)**.

Avogadro's number of formula units of any substance – that is, **one mole- has a mass in grams** numerically equal to the molecular weight of the substance.



Problem

Sulfur (S) is a nonmetallic element that is present in coal. When coal is burned, sulfur is converted to SO_2 and eventually to sulfuric acid that gives rise to acid rain phenomenon.

How many atoms are in 16.3 g of S?

Problem

Sulfur (S) is a nonmetallic element that is present in coal. When coal is burned, sulfur is converted to SO_2 and eventually to sulfuric acid that gives rise to acid rain phenomenon.

How many atoms are in 16.3 g of S?

Solve: $1 \text{ mol S} = 32.07 \text{ g S}$

We also know, $1 \text{ mol S} = 6.022 \times 10^{23} \text{ particles (atom)}$

$$16.3 \text{ g S} \times \frac{1 \text{ mol S}}{32.07 \text{ g S}} \times \frac{6.022 \times 10^{23} \text{ atom}}{1 \text{ mol S}} = 3.06 \times 10^{23} \text{ S atoms}$$

Mass and charge of subatomic particles

Table 2.1 Mass and Charge of Subatomic Particles

Particle	Mass (g)	Charge	
		Coulomb	Charge Unit
Electron*	9.10938×10^{-28}	-1.6022×10^{-19}	-1
Proton	1.67262×10^{-24}	$+1.6022 \times 10^{-19}$	+1
Neutron	1.67493×10^{-24}	0	0

Calculation of mass of electron

R. A. Millikan, an American physicist, found the charge of an electron to be $-1.6022 \times 10^{-19} \text{ C}$.

$$\begin{aligned}\text{mass of an electron} &= \frac{\text{charge}}{\text{charge/mass}} \\ &= \frac{-1.6022 \times 10^{-19} \text{ C}}{-1.76 \times 10^8 \text{ C/g}} \\ &= 9.10 \times 10^{-28} \text{ g}\end{aligned}$$